

commencis

Cloud Camp – Session 2

Virtual Private Cloud (VPC)

Amazon Web Services

- Region
- Availability Zone
- Local Zone
- Edge Locations

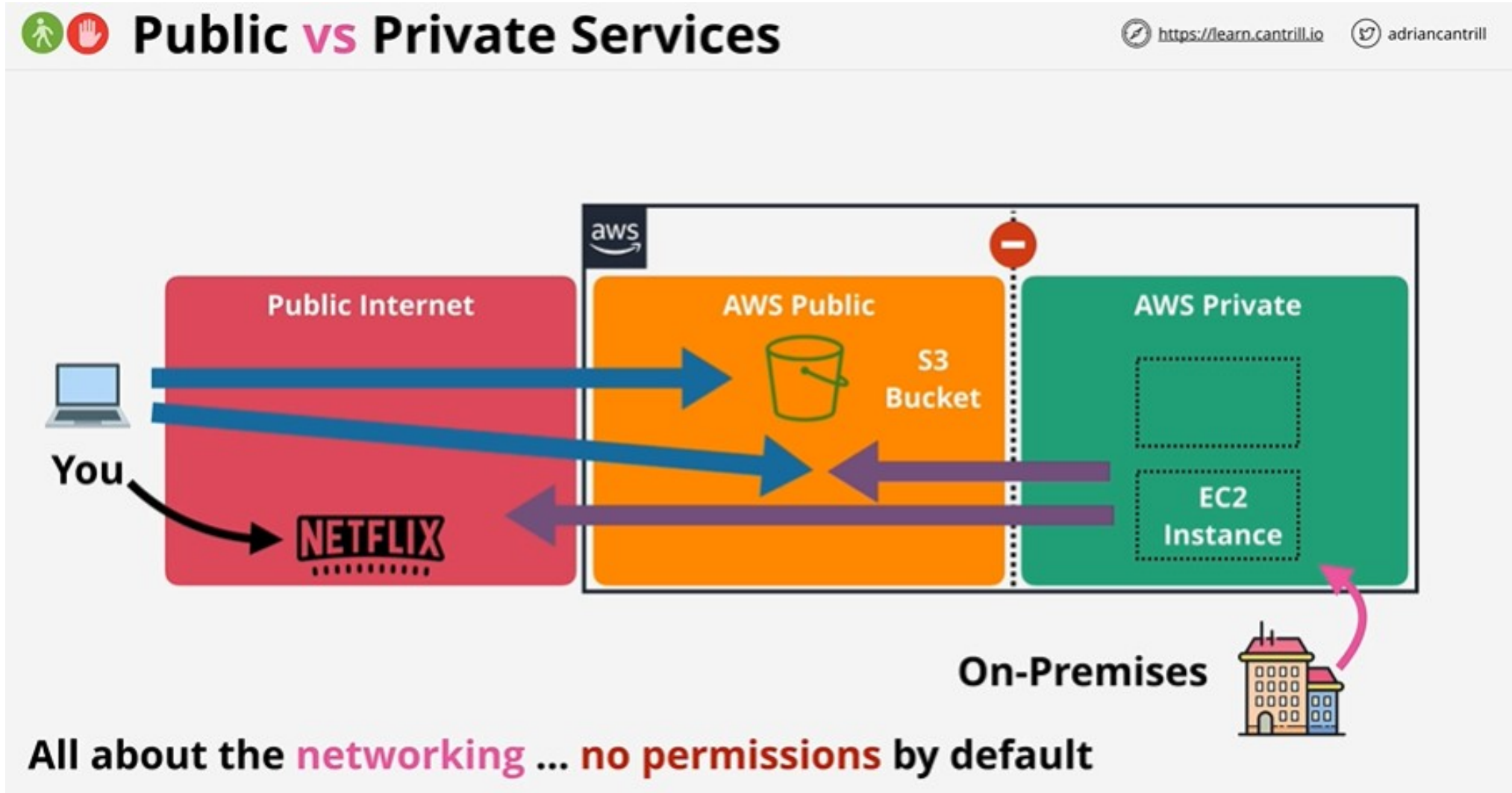




Virtual Private Cloud (VPC)

Logically isolated virtual network in the
AWS Cloud that you control.

commencis AWS Public vs Private Services



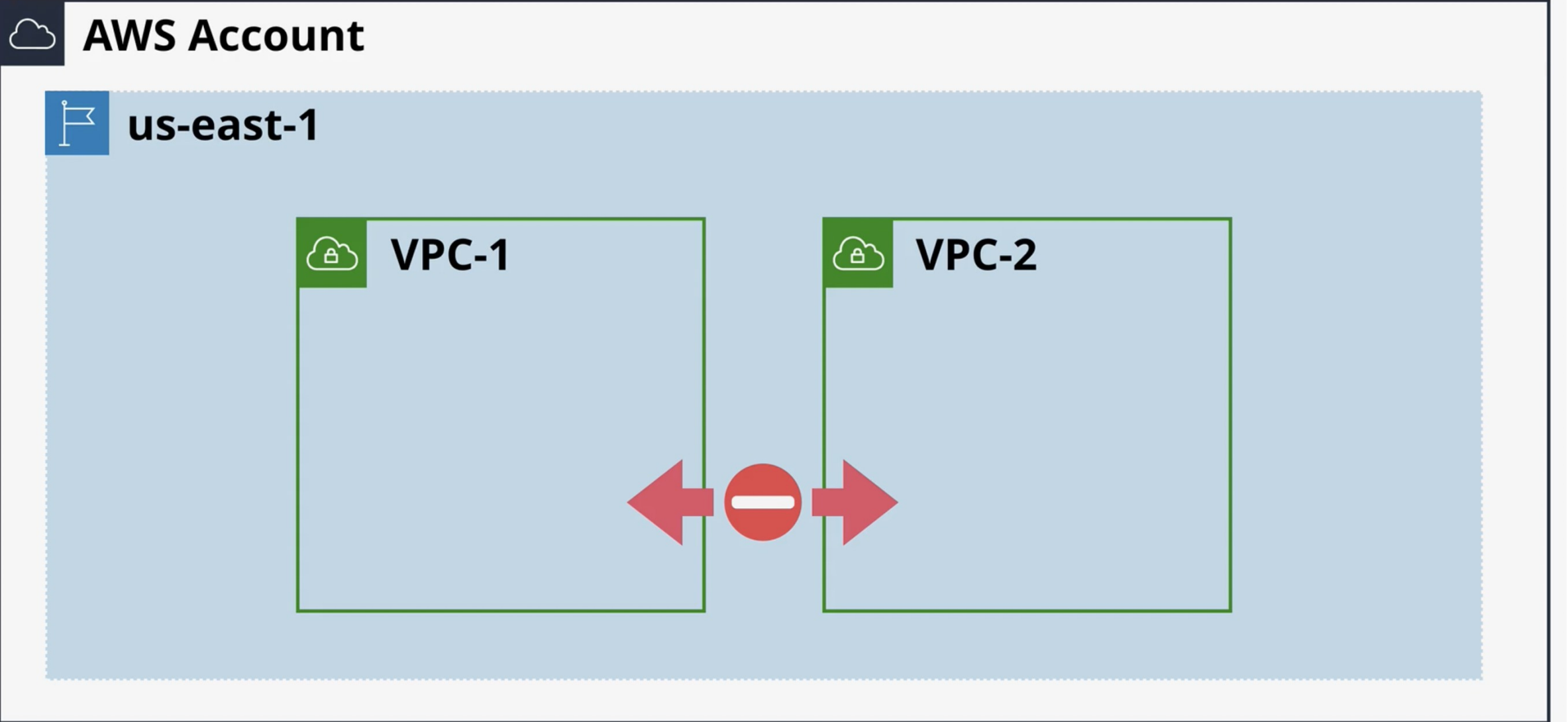
commencis AWS Global Infrastructure



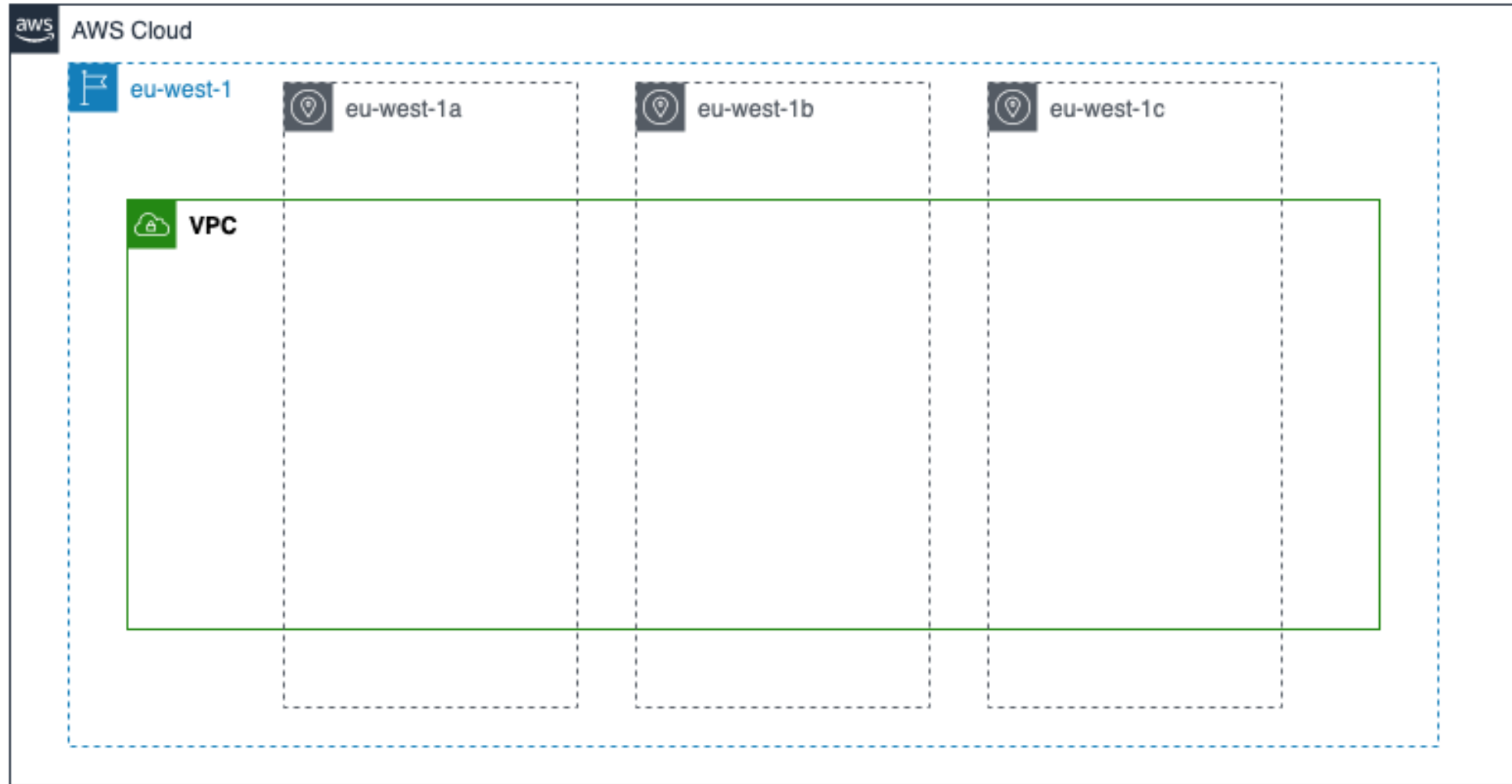
commencis VPC Basics

- A VPC is inside 1 account and 1 region.
- Spans all availability zones in a region.
- By default, private and isolated.
- If needed, a resource in a VPC can access and be accessed from the internet.

commencis VPC Basics



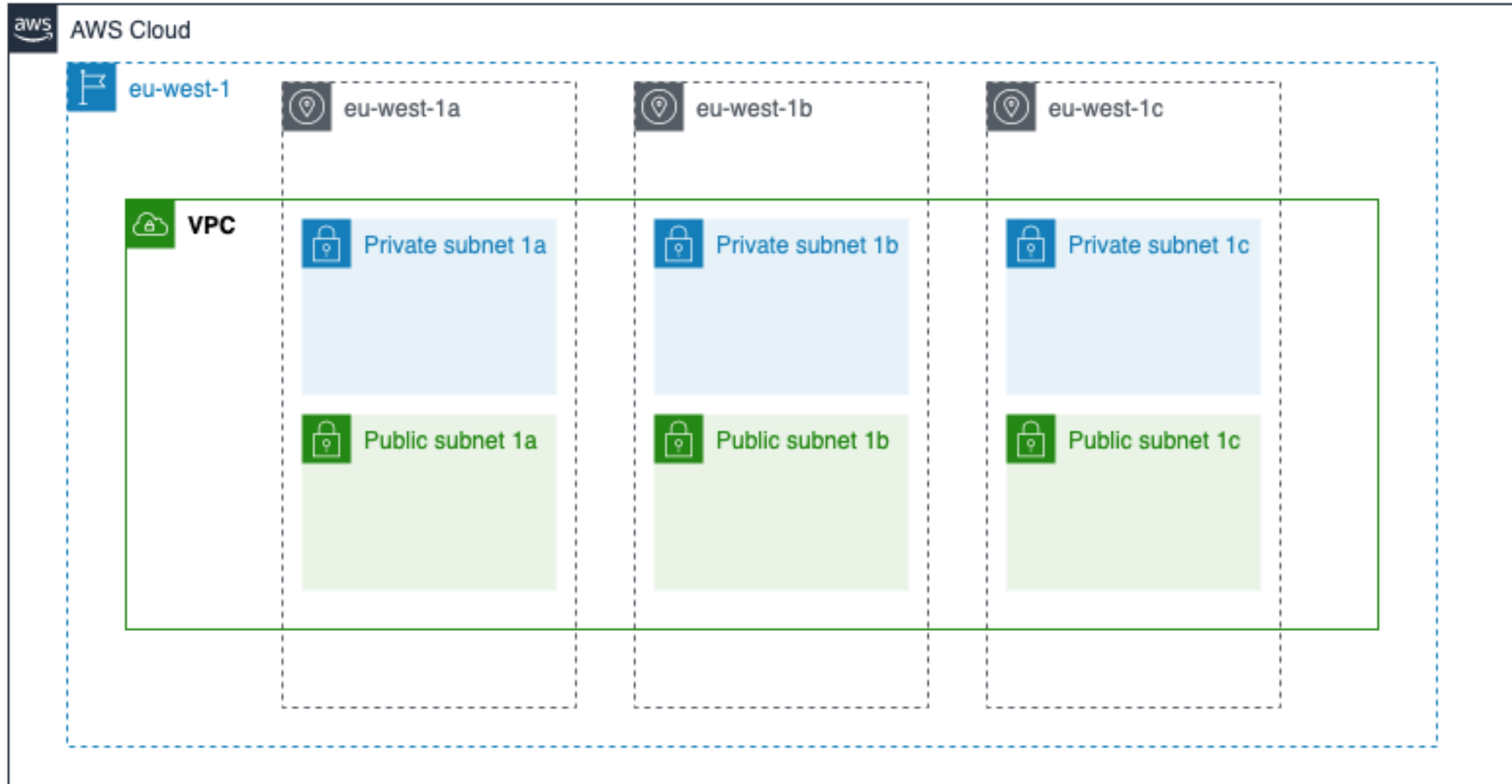
commencis VPC Basics



commencis VPC Components

- **Subnet:** A segment of a VPC in an AZ.
- **Internet Gateway (IGW):** The Amazon VPC side of a connection to the public internet. Gives resources access to the internet (two-way).
- **NAT Gateway (NGW):** Network Address Translation (NAT) service. Gives private subnet resources access to the internet (one-way).
- **Route Table:** A list of network routes attached to subnets.

commencis VPC Components - Subnet



commencis IP Ranges

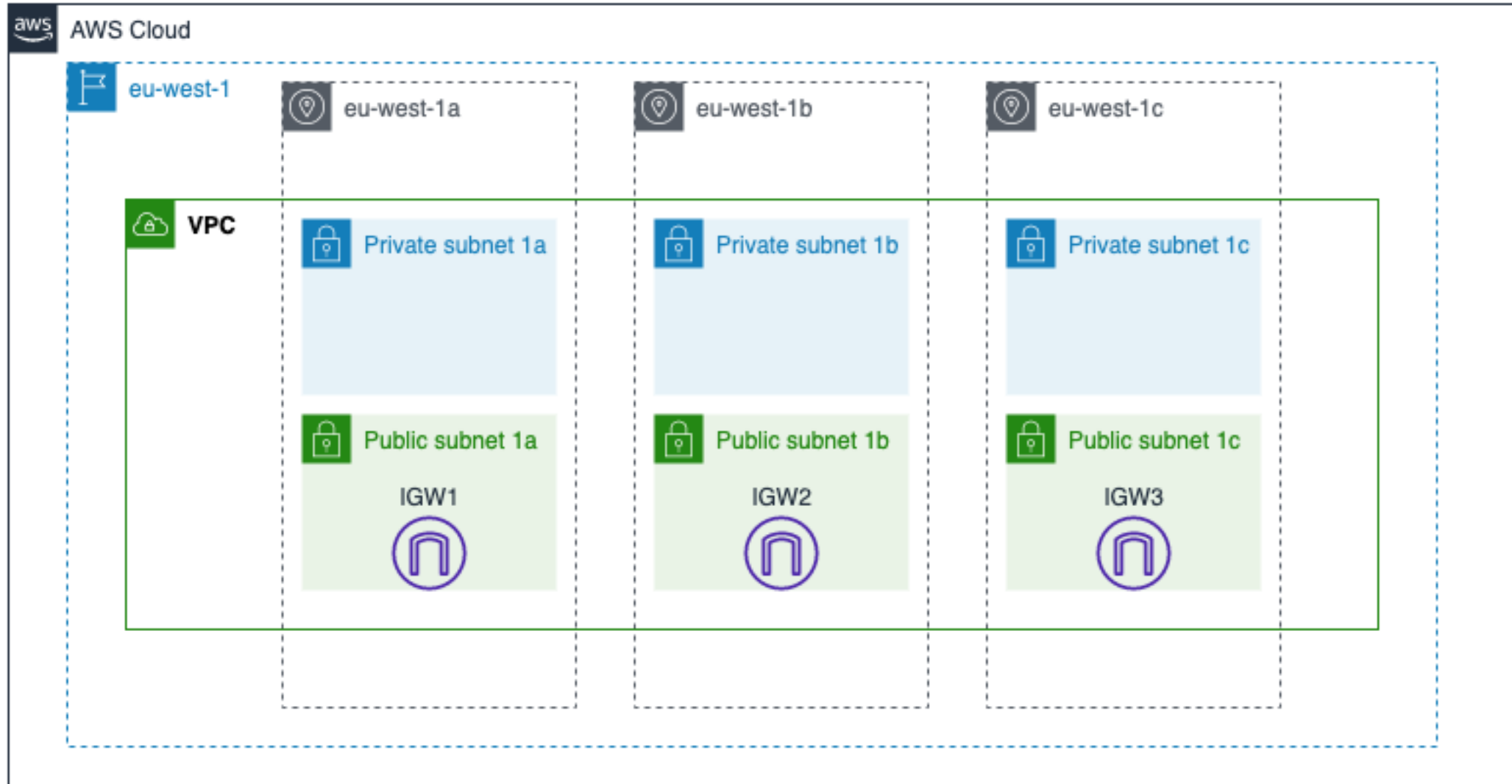
- A VPC needs to be assigned a private IP range when created.
- Subnets need to be assigned a subrange of its VPC's IP range.
- Private IP ranges can be in one of the following IP blocks:

Class	IP Range	CIDR notation
A	10.0.0.0 – 10.255.255.255	10.0.0.0/8
B	172.16.0.0 – 172.31.255.255	172.16.0.0/12
C	192.168.0.0 – 192.168.255.255	192.168.0.0/16

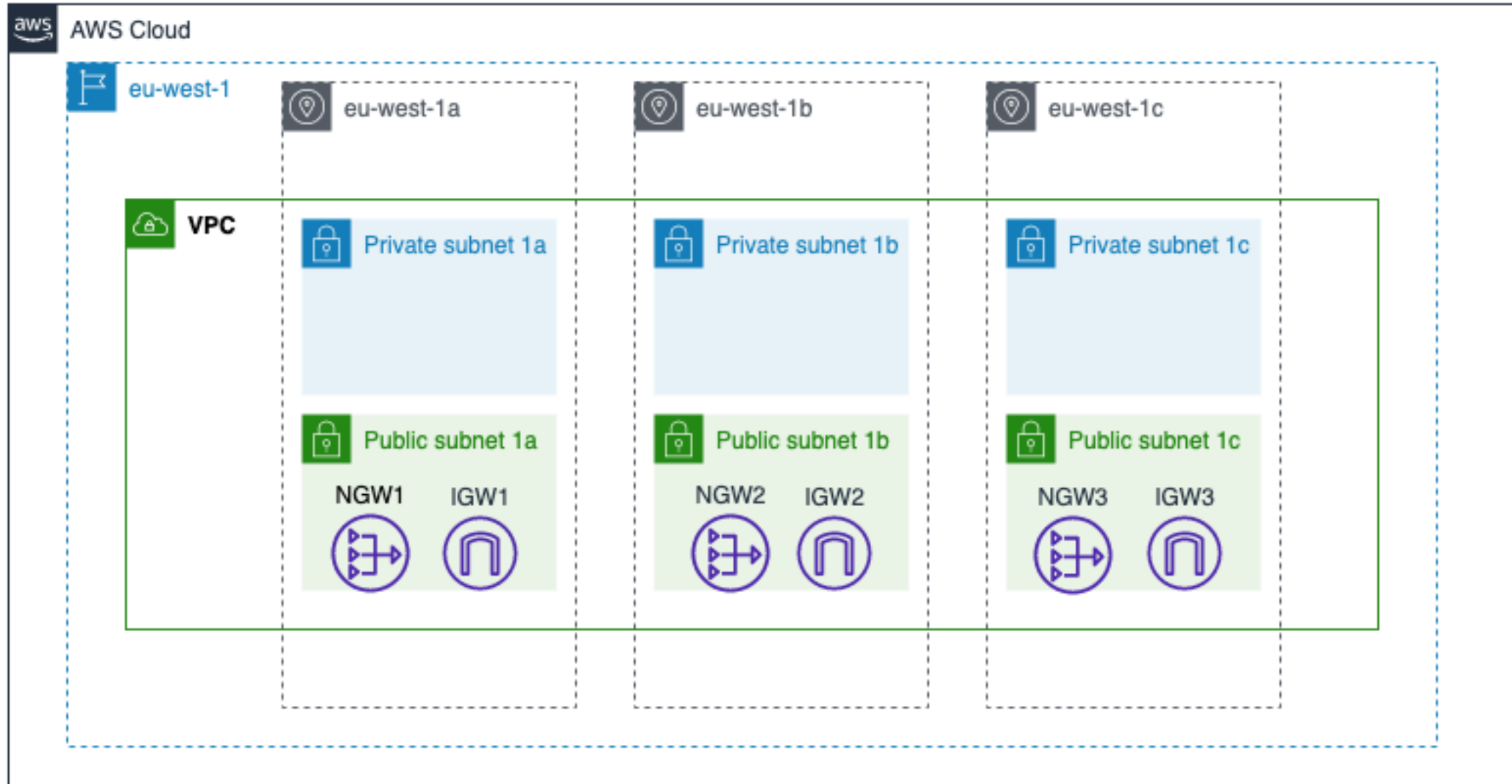
commencis IP Ranges - Example

- VPC has 10.10.0.0/16 IP range:
 - From 10.10.0.0 to 10.10.255.255
 - 65536 IP addresses
- Two private and two public subnets will be created, and each need about 200 IPs.
- We can assign:
 - Private-1: 10.10.0.0/24 => 10.10.0.0 – 10.10.0.255 => 256 IPs
 - Private-2: 10.10.1.0/24 => 10.10.1.0 – 10.10.1.255 => 256 IPs
 - Public-1: 10.10.2.0/24 => 10.10.2.0 – 10.10.2.255 => 256 IPs
 - Public-2: 10.10.3.0/24 => 10.10.3.0 – 10.10.3.255 => 256 Ips
 - 10.10.4.0 – 10.10.255.255 => Still available to use for future subnets

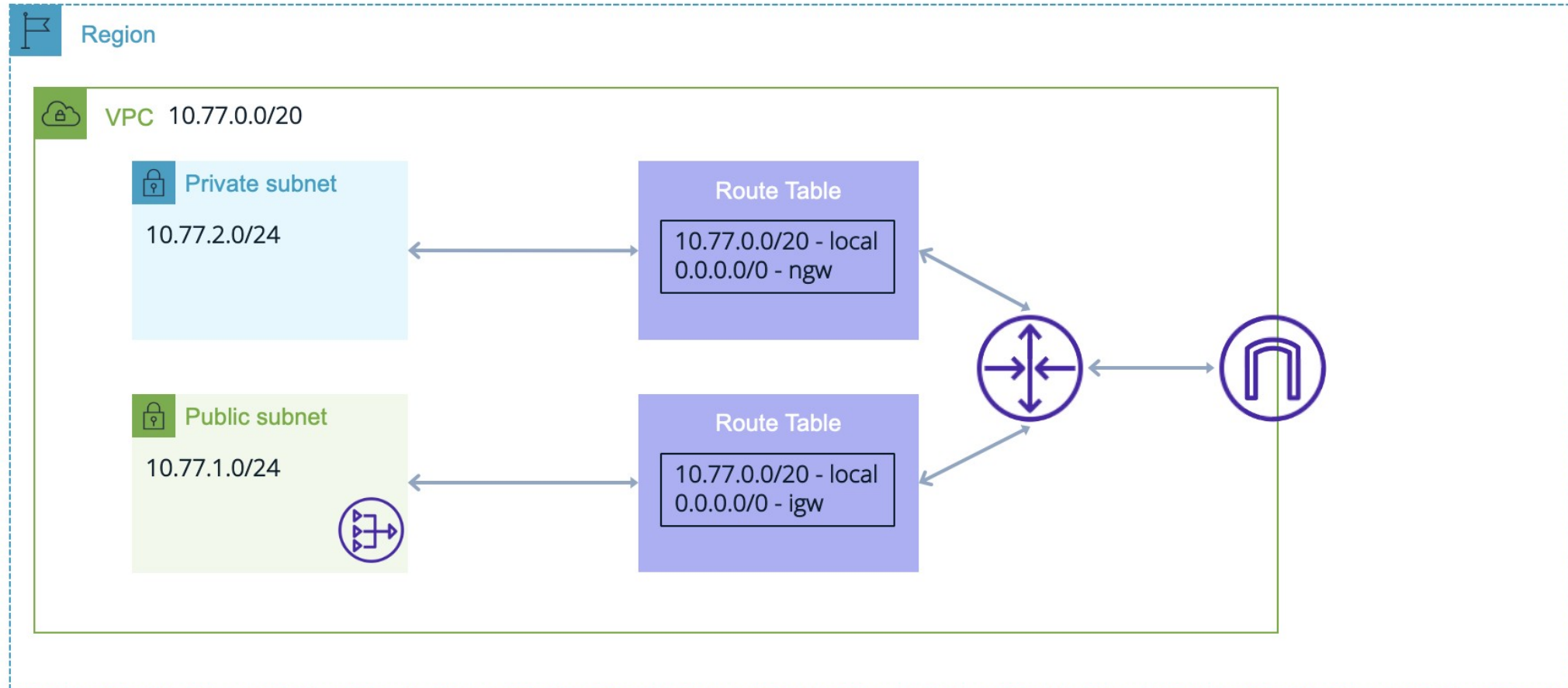
commencis VPC Components – Internet Gateway



commencis VPC Components – NAT Gateway



commencis VPC Components – Route Table



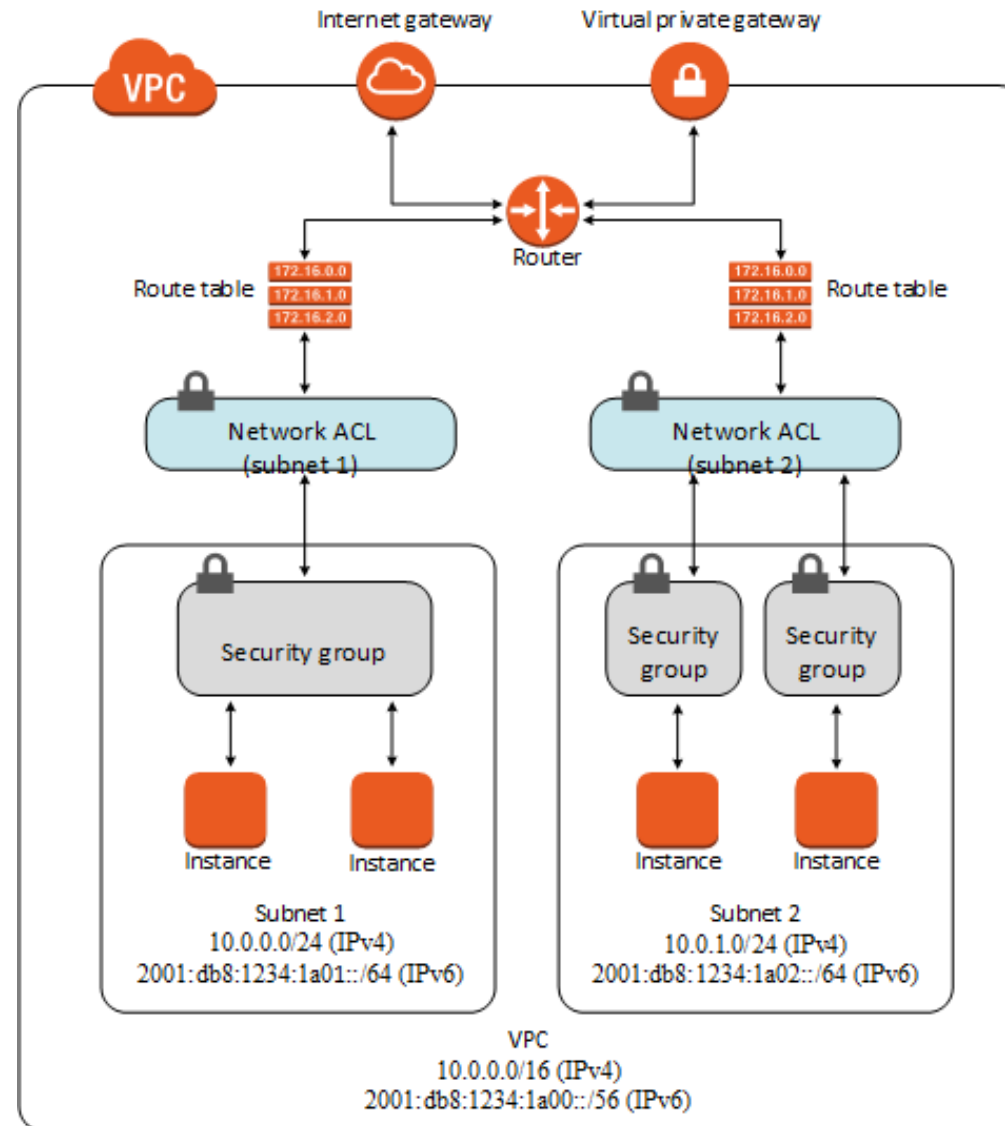
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VPC Setup Demo

commencis VPC Security

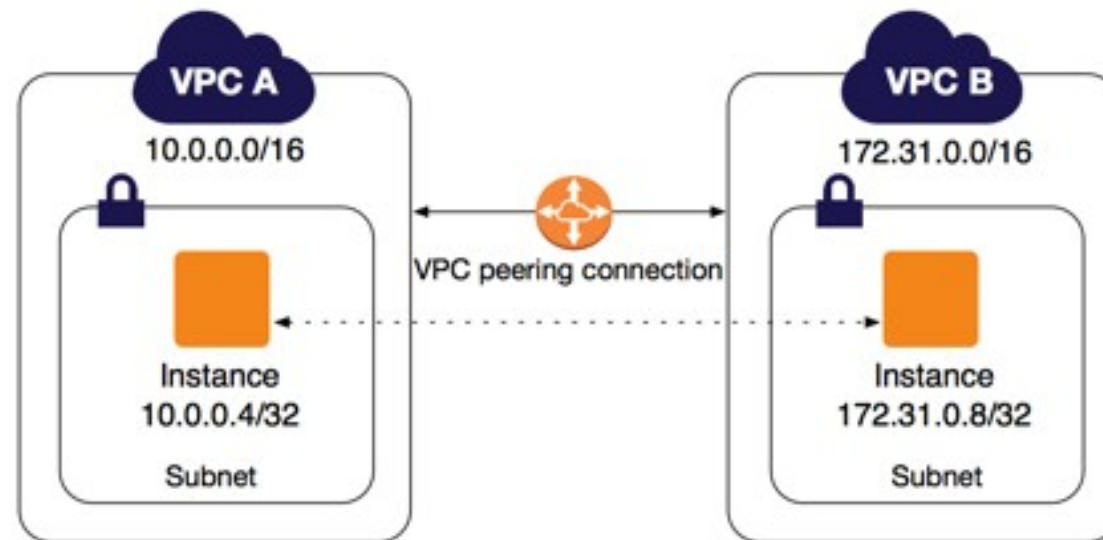
- VPC has two firewall mechanisms:
 - **Network Access Control List (NACL):**
 - Operates at subnet level.
 - Stateless firewall (request and response are seen as different).
 - Filters traffic going in and out of a subnet.
 - Doesn't impact connections that are in the subnet.
 - Can allow or deny traffic.
 - Can be used by multiple subnets.
 - **Security Group:**
 - Operates at resource level.
 - Stateful firewall (request and response are seen as together).
 - Filters traffic going in and out of a resource.
 - Can only allow traffic, not deny.
 - Can be used by multiple resources.

commencis VPC Security

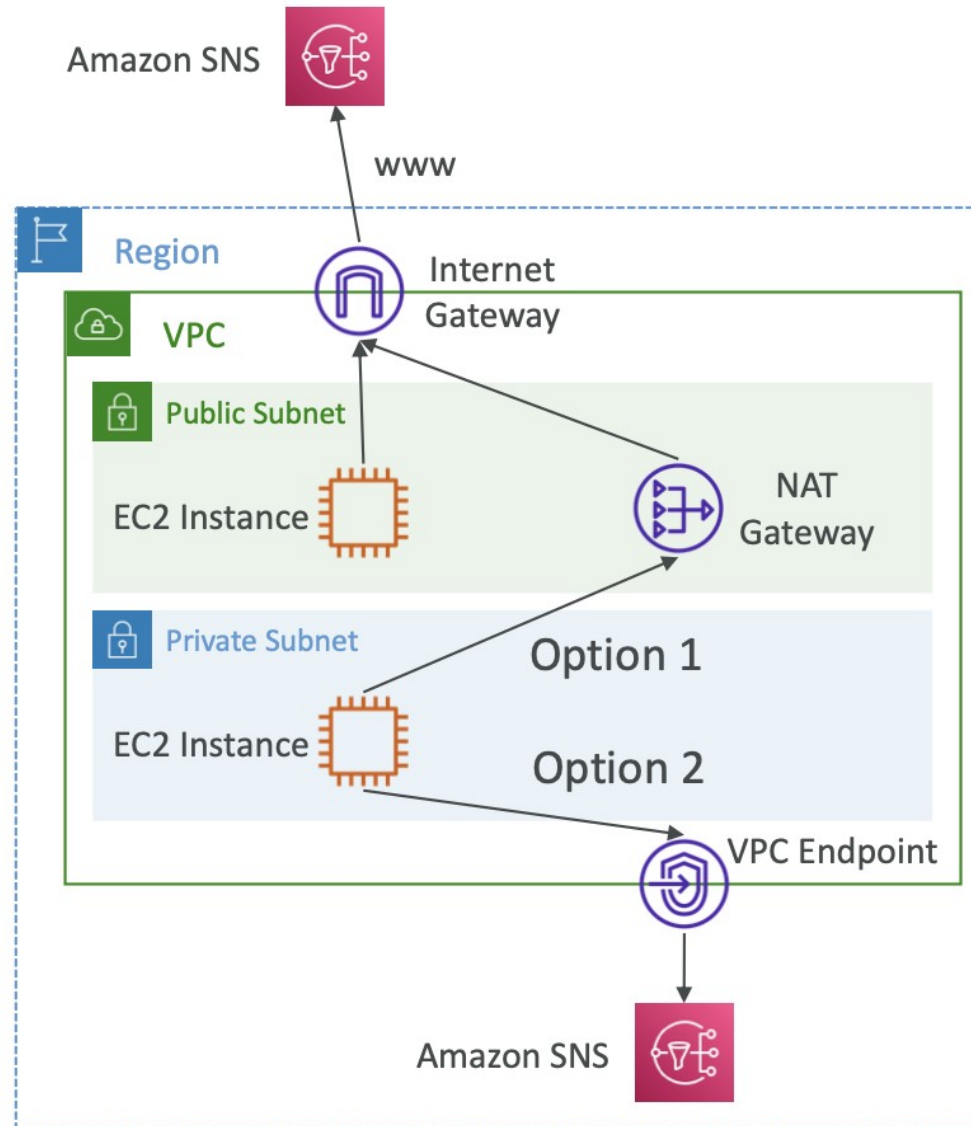


commencis VPC Peering

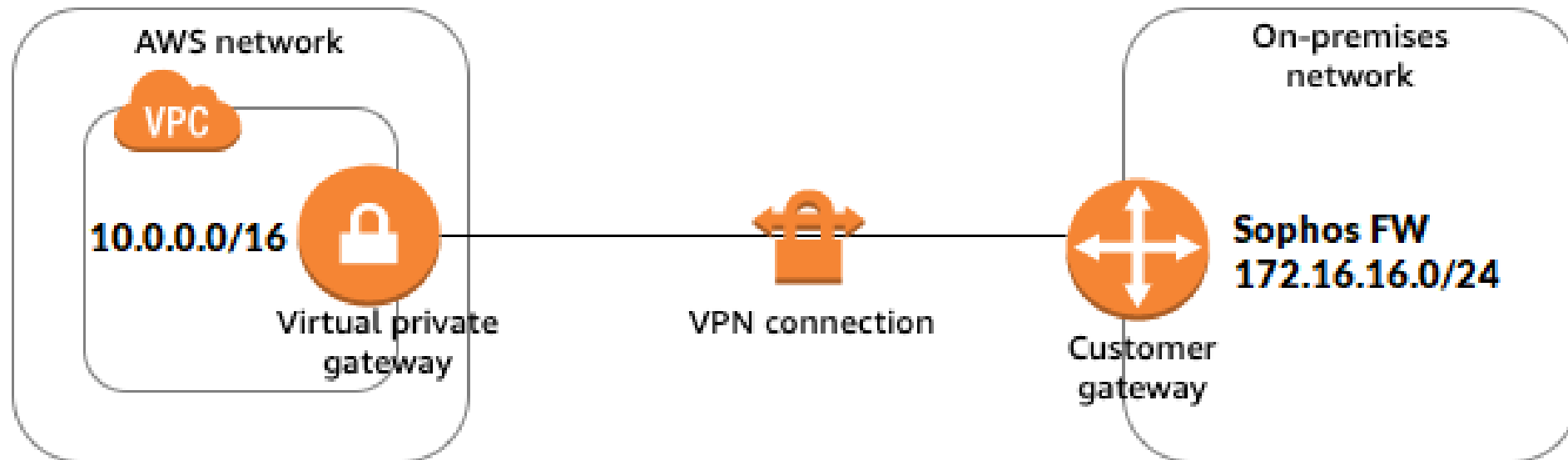
- Two VPC's cannot access each other by default.
- VPC peering creates a connection between two VPCs.
- It is possible to peer VPCs from different accounts.



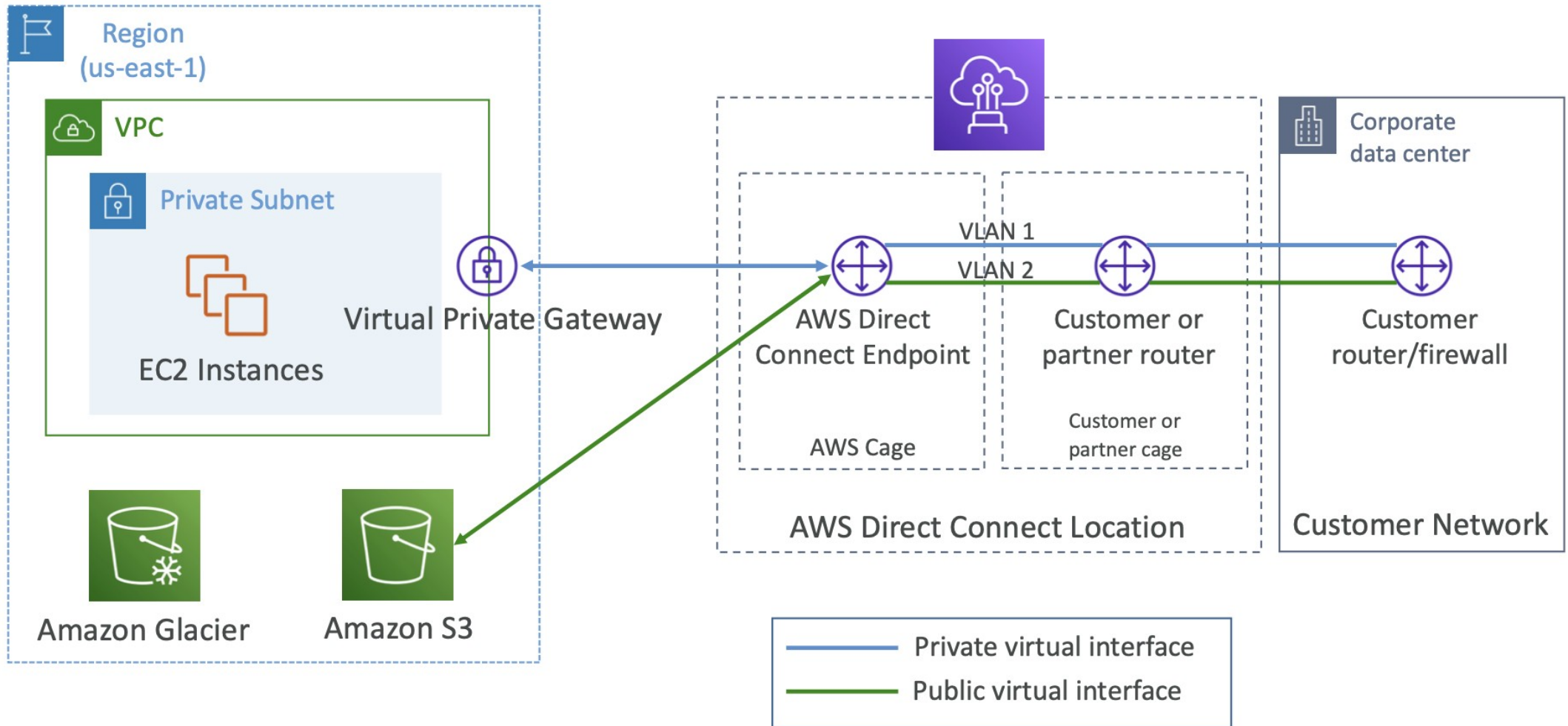
commencis VPC Endpoint (PrivateLink)



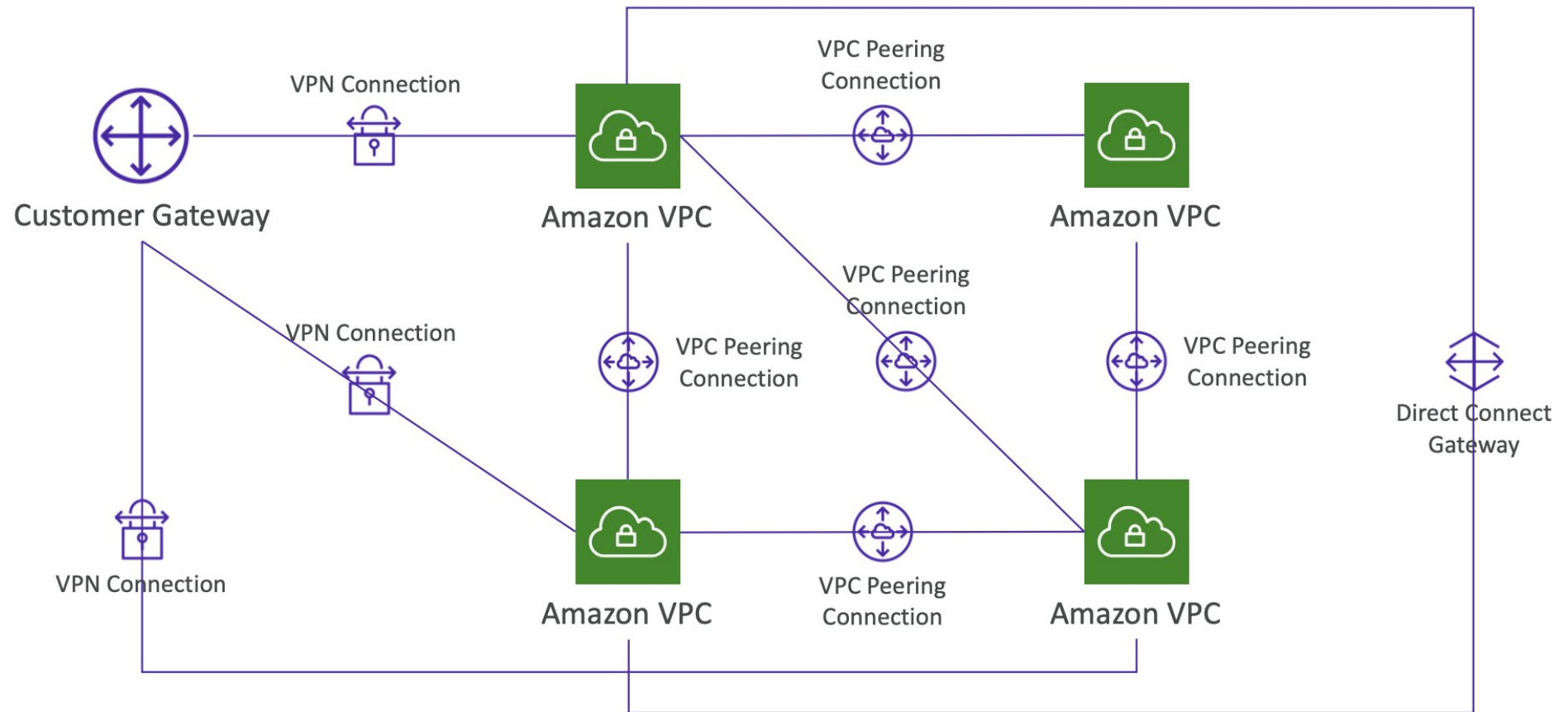
commencis AWS Site to Site VPN



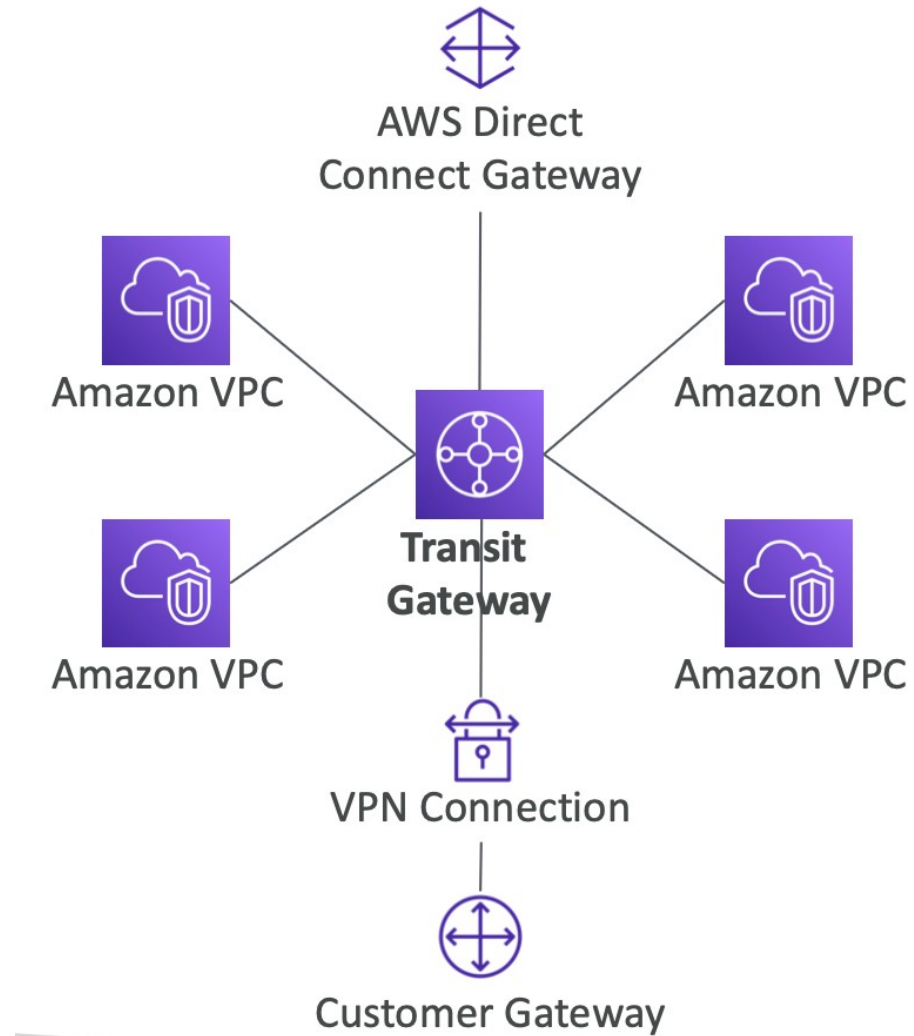
commencis Direct Connect



commencis Networking can get complex



commencis Transit Gateway



commencis Resources

- [Amazon Virtual Private Cloud Documentation](#)
- [Amazon Networking Fundamentals @ Youtube](#)
- [Terraform VPC module](#)
- [Adrian Cantrill's AWS Developer course](#)
- [Stephane Maarek's AWS Solutions Architect course](#)

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Thank you.